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Experiment 5

Experiment 5: Hadoop Installation, Configuration, and HDFS Operations

Aim: To install and configure Hadoop in a single-node environment and perform basic HDFS operations using command-line tools.

Mapped CO(s): CO2

1. Import Required Libraries

In [1]:
```
import os
import shutil
import subprocess
import time
from pathlib import Path
from shlex import quote
```

2. Hadoop Ecosystem (Brief Theory)

Apache Hadoop is an open-source framework for storing and processing large datasets in a distributed, fault-tolerant way.

Component	Role
HDFS	Hadoop Distributed File System — stores data as blocks across nodes
YARN	Resource manager and job scheduler
MapReduce	Parallel processing model
Hadoop Common	Shared utilities and libraries

This experiment uses a **single-node (pseudo-distributed)** setup: NameNode and DataNode run on the same machine, with replication factor **1**.

Hadoop 3.4.1 is installed at `/Users/ojas/hadoop-3.4.1` (a path without spaces). Java 17 is used from Homebrew OpenJDK.

3. Set Java and Hadoop Environment

In [2]:
```
EXP_DIR = Path("/Users/ojas/Desktop/DS Lab/Experiment5")
LOCAL_DIR = Path("/tmp/exp5-local")
DOWNLOAD_DIR = Path("/tmp/exp5-downloads")
HDFS_DATA = Path("/tmp/exp5-hdfs")
CONF_DIR = Path("/tmp/exp5-hadoop-conf")
LOG_DIR = Path("/tmp/exp5-hadoop-logs")

JAVA_HOME = Path("/opt/homebrew/opt/openjdk@17/libexec/openjdk.jdk/Conten
```

```
HADOOP_HOME = Path("/Users/ojas/hadoop-3.4.1")

os.environ["JAVA_HOME"] = str(JAVA_HOME)
os.environ["HADOOP_HOME"] = str(HADOOP_HOME)
os.environ["HADOOP_CONF_DIR"] = str(CONF_DIR)
os.environ["HADOOP_LOG_DIR"] = str(LOG_DIR)
os.environ["PATH"] = (
    f"{JAVA_HOME / 'bin'}:{HADOOP_HOME / 'bin'}:{HADOOP_HOME / 'sbin'}:"
    + os.environ["PATH"]
)

for folder in (LOG_DIR, HDFS_DATA, DOWNLOAD_DIR, LOCAL_DIR):
    folder.mkdir(parents=True, exist_ok=True)

for src in (EXP_DIR / "local_files").glob("*.txt"):
    shutil.copy2(src, LOCAL_DIR / src.name)

if not (CONF_DIR / "log4j.properties").exists():
    if CONF_DIR.exists():
        shutil.rmtree(CONF_DIR)
    shutil.copytree(HADOOP_HOME / "etc" / "hadoop", CONF_DIR)


def run(cmd, check=True, quiet=False):
    if not quiet:
        print(f"$ {cmd}")
    result = subprocess.run(
        cmd,
        shell=True,
        text=True,
        capture_output=True,
        env=os.environ.copy(),
    )
    out = (result.stdout or "").rstrip()
    err = (result.stderr or "").rstrip()
    if out:
        print(out)
    if err:
        err = "\n".join(
            line for line in err.splitlines()
            if "NativeCodeLoader" not in line
        )
        if err:
            print(err)
    if check and result.returncode != 0:
        raise RuntimeError(f"Command failed with exit code {result.return
    return result

print("JAVA_HOME      :", os.environ["JAVA_HOME"])
print("HADOOP_HOME    :", os.environ["HADOOP_HOME"])
print("HADOOP_CONF_DIR:", os.environ["HADOOP_CONF_DIR"])
```

```
JAVA_HOME      : /opt/homebrew/opt/openjdk@17/libexec/openjdk.jdk/Content
s/Home
HADOOP_HOME    : /Users/ojas/hadoop-3.4.1
HADOOP_CONF_DIR: /tmp/exp5-hadoop-conf
```

4. Check Java and Hadoop Versions

In [3]:
```
run("java -version")
print()
run("hadoop version")
```

```
$ java -version
openjdk version "17.0.19" 2026-04-21
OpenJDK Runtime Environment Homebrew (build 17.0.19+0)
OpenJDK 64-Bit Server VM Homebrew (build 17.0.19+0, mixed mode, sharing)

$ hadoop version
Hadoop 3.4.1
Source code repository https://github.com/apache/hadoop.git -r 4d782530934
8956336b8f06a08322b78422849b1
Compiled by mthakur on 2024-10-09T14:57Z
Compiled on platform linux-x86_64
Compiled with protoc 3.23.4
From source with checksum 7292fe9dba5e2e44e3a9f763fce3e680
This command was run using /Users/ojas/hadoop-3.4.1/share/hadoop/common/ha
doop-common-3.4.1.jar
```

Out[3]:
```
CompletedProcess(args='hadoop version', returncode=0, stdout='Hadoop 3.
4.1\nSource code repository https://github.com/apache/hadoop.git -r 4d78
25309348956336b8f06a08322b78422849b1\nCompiled by mthakur on 2024-10-09T
14:57Z\nCompiled on platform linux-x86_64\nCompiled with protoc 3.23.4\n
From source with checksum 7292fe9dba5e2e44e3a9f763fce3e680\nThis command
was run using /Users/ojas/hadoop-3.4.1/share/hadoop/common/hadoop-common
-3.4.1.jar\n', stderr='')
```

5. Configure Hadoop XML Files

Default Hadoop configuration is copied, then these properties are set:

- `core-site.xml` : `fs.defaultFS = hdfs://localhost:9000`
- `hdfs-site.xml` : replication = 1, NameNode and DataNode directories

In [4]:
```
default_conf = HADOOP_HOME / "etc" / "hadoop"
if CONF_DIR.exists():
    shutil.rmtree(CONF_DIR)
shutil.copytree(default_conf, CONF_DIR)

env_sh = CONF_DIR / "hadoop-env.sh"
env_text = env_sh.read_text()
if "EXP5 JAVA_HOME" not in env_text:
    env_sh.write_text(
        env_text + f"\n# EXP5 JAVA_HOME\nexport JAVA_HOME={JAVA_HOME}\n"
    )

(CONF_DIR / "core-site.xml").write_text("""<?xml version="1.0" encoding="
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
""")

(CONF_DIR / "hdfs-site.xml").write_text(f"""<?xml version="1.0" encoding=
```

```
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file://{HDFS_DATA / 'namenode'}</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file://{HDFS_DATA / 'datanode'}</value>
  </property>
  <property>
    <name>dfs.namenode.http-address</name>
    <value>localhost:9870</value>
  </property>
  <property>
    <name>dfs.datanode.address</name>
    <value>localhost:9866</value>
  </property>
  <property>
    <name>dfs.datanode.http.address</name>
    <value>localhost:9864</value>
  </property>
  <property>
    <name>dfs.datanode.ipc.address</name>
    <value>localhost:9867</value>
  </property>
  <property>
    <name>dfs.permissions.enabled</name>
    <value>false</value>
  </property>
</configuration>
""")

print("Configuration written to", CONF_DIR)
print("\n--- core-site.xml ---")
print((CONF_DIR / "core-site.xml").read_text())
print("--- hdfs-site.xml ---")
print((CONF_DIR / "hdfs-site.xml").read_text())
```

```
Configuration written to /tmp/exp5-hadoop-conf

--- core-site.xml ---
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>

--- hdfs-site.xml ---
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///tmp/exp5-hdfs/namenode</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///tmp/exp5-hdfs/datanode</value>
  </property>
  <property>
    <name>dfs.namenode.http-address</name>
    <value>localhost:9870</value>
  </property>
  <property>
    <name>dfs.datanode.address</name>
    <value>localhost:9866</value>
  </property>
  <property>
    <name>dfs.datanode.http.address</name>
    <value>localhost:9864</value>
  </property>
  <property>
    <name>dfs.datanode.ipc.address</name>
    <value>localhost:9867</value>
  </property>
  <property>
    <name>dfs.permissions.enabled</name>
    <value>false</value>
  </property>
</configuration>
```

6. Format the Hadoop NameNode

In [5]:

```
run("hdfs --daemon stop datanode", check=False, quiet=True)
run("hdfs --daemon stop namenode", check=False, quiet=True)
time.sleep(2)

if HDFS_DATA.exists():
    shutil.rmtree(HDFS_DATA)
```

```
HDFS_DATA.mkdir(parents=True, exist_ok=True)

run("hdfs namenode -format -force -nonInteractive")
```

```
$ hdfs namenode -format -force -nonInteractive
2026-09-13 23:47:20 INFO  NameNode:809 - STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = Ojass-MacBook-Air.local/127.0.0.1
STARTUP_MSG:   args = [-format, -force, -nonInteractive]
STARTUP_MSG:   version = 3.4.1
STARTUP_MSG:   classpath = /tmp/exp5-hadoop-conf:/Users/ojas/hadoop-3.4.1/
share/hadoop/common/lib/jaxb-impl-2.2.3-1.jar:/Users/ojas/hadoop-3.4.1/sha
re/hadoop/common/lib/jetty-webapp-9.4.53.v20231009.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/common/lib/curator-client-5.2.0.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/common/lib/netty-handler-4.1.100.Final.jar:/Users/ojas/
hadoop-3.4.1/share/hadoop/common/lib/kerby-pkix-2.0.3.jar:/Users/ojas/hado
op-3.4.1/share/hadoop/common/lib/commons-logging-1.2.jar:/Users/ojas/hadoo
p-3.4.1/share/hadoop/common/lib/reload4j-1.2.22.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/common/lib/jackson-core-2.12.7.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/common/lib/netty-all-4.1.100.Final.jar:/Users/ojas/hadoop
-3.4.1/share/hadoop/common/lib/netty-codec-redis-4.1.100.Final.jar:/Users/
ojas/hadoop-3.4.1/share/hadoop/common/lib/jersey-json-1.22.0.jar:/Users/oj
as/hadoop-3.4.1/share/hadoop/common/lib/slf4j-api-1.7.36.jar:/Users/ojas/h
adoop-3.4.1/share/hadoop/common/lib/netty-codec-dns-4.1.100.Final.jar:/Use
rs/ojas/hadoop-3.4.1/share/hadoop/common/lib/dnsjava-3.6.1.jar:/Users/oja
s/hadoop-3.4.1/share/hadoop/common/lib/jetty-util-ajax-9.4.53.v20231009.ja
r:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-stomp-4.1.1
00.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-comp
ress-1.26.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/failureac
cess-1.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/checker-qual
-2.5.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jersey-servlet
-1.19.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/slf4j-reload4
j-1.7.36.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transp
ort-native-epoll-4.1.100.Final-linux-aarch_64.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/common/lib/kerby-asn1-2.0.3.jar:/Users/ojas/hadoop-3.4.1/sh
are/hadoop/common/lib/commons-cli-1.5.0.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/common/lib/woodstox-core-5.4.0.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/common/lib/zookeeper-jute-3.8.4.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/common/lib/curator-framework-5.2.0.jar:/Users/ojas/hadoop-3.4.1/s
hare/hadoop/common/lib/httpcore-4.4.13.jar:/Users/ojas/hadoop-3.4.1/share/
hadoop/common/lib/kerby-config-2.0.3.jar:/Users/ojas/hadoop-3.4.1/share/ha
doop/common/lib/netty-codec-smtp-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/common/lib/kerb-util-2.0.3.jar:/Users/ojas/hadoop-3.4.1/sha
re/hadoop/common/lib/hadoop-shaded-guava-1.3.0.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/common/lib/netty-codec-mqtt-4.1.100.Final.jar:/Users/ojas/h
adoop-3.4.1/share/hadoop/common/lib/zookeeper-3.8.4.jar:/Users/ojas/hadoop
-3.4.1/share/hadoop/common/lib/commons-net-3.9.0.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/common/lib/hadoop-annotations-3.4.1.jar:/Users/ojas/hadoo
p-3.4.1/share/hadoop/common/lib/javax.servlet-api-3.1.0.jar:/Users/ojas/ha
doop-3.4.1/share/hadoop/common/lib/jersey-server-1.19.4.jar:/Users/ojas/ha
doop-3.4.1/share/hadoop/common/lib/commons-io-2.16.1.jar:/Users/ojas/hadoo
p-3.4.1/share/hadoop/common/lib/hadoop-shaded-protobuf_3_25-1.3.0.jar:/Use
rs/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-server-9.4.53.v2023100
9.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-util-9.4.53.v
20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transp
ort-rxtx-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/li
b/guava-27.0-jre.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/nett
y-codec-http-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/commo
n/lib/netty-buffer-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoo
p/common/lib/gson-2.9.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/l
ib/jcip-annotations-1.0-1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/commo
n/lib/netty-common-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoo
p/common/lib/jetty-io-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/
```

hadoop/common/lib/jsr311-api-1.1.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/stax2-api-4.2.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-haproxy-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-socks-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-classes-epoll-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/hadoop-auth-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/nimbus-jose-jwt-9.37.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-xml-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-handler-ssl-ocsp-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-handler-proxy-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jaxb-api-2.2.11.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-resolver-dns-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jakarta.activation-api-1.2.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/snappy-java-1.1.10.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-resolver-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/listenablefuture-9999.0-empty-to-avoid-conflict-with-guava.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/metrics-core-3.2.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jsp-api-2.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/avro-1.9.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/kerb-core-2.0.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jersey-core-1.19.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-math3-3.6.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-native-kqueue-4.1.100.Final-osx-aarch_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-resolver-dns-native-macos-4.1.100.Final-osx-aarch_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-sctp-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-native-unix-common-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/j2objc-annotations-1.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jul-to-slf4j-1.7.36.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-classes-kqueue-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-servlet-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-collections-3.2.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/kerby-util-2.0.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-codec-1.15.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jettison-1.5.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/bcprov-jdk18on-1.78.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/audience-annotations-0.12.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-resolver-dns-native-macos-4.1.100.Final-osx-x86_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/curator-recipes-5.2.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-resolver-dns-classes-macos-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-beanutils-1.9.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-native-epoll-4.1.100.Final-linux-x86_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/kerb-crypto-2.0.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jackson-annotations-2.12.7.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jsch-0.1.55.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-udt-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-security-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jackson-databind-2.12.7.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-memcache-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/re2j-1.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-configuration2-2.10.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-lang3-3.12.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/httpclient-4.5.13.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/animal-sn

```
iffer-annotations-1.17.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/li
b/netty-transport-native-epoll-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/
share/hadoop/common/lib/jetty-http-9.4.53.v20231009.jar:/Users/ojas/hadoop
-3.4.1/share/hadoop/common/lib/jsr305-3.0.2.jar:/Users/ojas/hadoop-3.4.1/s
hare/hadoop/common/lib/netty-codec-http2-4.1.100.Final.jar:/Users/ojas/had
oop-3.4.1/share/hadoop/common/lib/commons-daemon-1.0.13.jar:/Users/ojas/ha
doop-3.4.1/share/hadoop/common/lib/jetty-xml-9.4.53.v20231009.jar:/Users/o
jas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-native-kqueue-4.
1.100.Final-osx-x86_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/li
b/commons-text-1.10.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/had
oop-nfs-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/hadoop-kms-
3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/hadoop-registry-3.
4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/hadoop-common-3.4.1-t
ests.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/hadoop-common-3.4.1.
jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs:/Users/ojas/hadoop-3.4.1/sh
are/hadoop/hdfs/lib/jaxb-impl-2.2.3-1.jar:/Users/ojas/hadoop-3.4.1/share/h
adoop/hdfs/lib/jetty-webapp-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/
share/hadoop/hdfs/lib/curator-client-5.2.0.jar:/Users/ojas/hadoop-3.4.1/sh
are/hadoop/hdfs/lib/netty-handler-4.1.100.Final.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/hdfs/lib/kerby-pkix-2.0.3.jar:/Users/ojas/hadoop-3.4.1/sh
are/hadoop/hdfs/lib/commons-logging-1.2.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/hdfs/lib/reload4j-1.2.22.jar:/Users/ojas/hadoop-3.4.1/share/hadoo
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ers/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/curator-framework-5.2.0.jar:/U
sers/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/httpcore-4.4.13.jar:/Users/oj
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adoop-3.4.1/share/hadoop/hdfs/lib/netty-codec-smtp-4.1.100.Final.jar:/User
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jas/hadoop-3.4.1/share/hadoop/hdfs/lib/commons-net-3.9.0.jar:/Users/ojas/h
adoop-3.4.1/share/hadoop/hdfs/lib/hadoop-annotations-3.4.1.jar:/Users/oja
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ers/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-resolver-dns-4.1.100.Fin
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adoop/hdfs/lib/netty-resolver-dns-classes-macos-4.1.100.Final.jar:/Users/o
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s/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jsr305-3.0.2.jar:/Users/ojas/had
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re/hadoop/mapreduce/hadoop-mapreduce-client-jobclient-3.4.1-tests.jar:/Use
rs/ojas/hadoop-3.4.1/share/hadoop/mapreduce/hadoop-mapreduce-client-jobcli
ent-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/mapreduce/hadoop-mapre
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e/hadoop/mapreduce/hadoop-mapreduce-client-hs-plugins-3.4.1.jar:/Users/oja
s/hadoop-3.4.1/share/hadoop/mapreduce/hadoop-mapreduce-client-hs-3.4.1.ja
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ce/hadoop-mapreduce-client-core-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/h
adoop/yarn:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/javax.websocket-
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2.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/javax.activation-ap
i-1.2.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/codemodel-2.6.j
ar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/asm-commons-9.6.jar:/Use
rs/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jackson-jaxrs-json-provider-2.1
2.7.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/objenesis-2.6.jar:/
Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jline-3.9.0.jar:/Users/ojas/
hadoop-3.4.1/share/hadoop/yarn/lib/ehcache-3.8.2.jar:/Users/ojas/hadoop-3.
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s/hadoop-3.4.1/share/hadoop/yarn/lib/asm-tree-9.6.jar:/Users/ojas/hadoop-
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4.1/share/hadoop/yarn/lib/websocket-client-9.4.53.v20231009.jar:/Users/oja
s/hadoop-3.4.1/share/hadoop/yarn/lib/jsonschema2pojo-core-1.0.2.jar:/User
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31009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/javax-websocket-c
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ib/jackson-module-jaxb-annotations-2.12.7.jar:/Users/ojas/hadoop-3.4.1/sha
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adoop/yarn/hadoop-yarn-server-nodemanager-3.4.1.jar:/Users/ojas/hadoop-3.
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yarn-server-timeline-pluginstorage-3.4.1.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/yarn/hadoop-yarn-applications-distributedshell-3.4.1.jar:/Users/o
jas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-globalpolicygenerato
r-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-
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3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-api-3.4.
1.jar
STARTUP_MSG:   build = https://github.com/apache/hadoop.git -r 4d782530934
8956336b8f06a08322b78422849b1; compiled by 'mthakur' on 2024-10-09T14:57Z
STARTUP_MSG:   java = 17.0.19
************************************************************/
2026-09-13 23:47:20 INFO  NameNode:91 - registered UNIX signal handlers fo
r [TERM, HUP, INT]
2026-09-13 23:47:20 INFO  NameNode:1846 - createNameNode [-format, -force,
-nonInteractive]
2026-09-13 23:47:20 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
2026-09-13 23:47:20 INFO  NameNode:1393 - Formatting using clusterid: CID-
30f97f04-b1e6-4a5c-846c-7e0c79163e13
2026-09-13 23:47:20 INFO  FSEditLog:238 - Edit logging is async:true
2026-09-13 23:47:20 INFO  FSNamesystem:869 - KeyProvider: null
2026-09-13 23:47:20 INFO  FSNamesystem:142 - fsLock is fair: true
2026-09-13 23:47:20 INFO  FSNamesystem:160 - Detailed lock hold time metri
```

```
cs enabled: false
2026-09-13 23:47:20 INFO  FSNamesystem:904 - fsOwner                = ojas
(auth:SIMPLE)
2026-09-13 23:47:20 INFO  FSNamesystem:905 - supergroup             = supe
rgroup
2026-09-13 23:47:20 INFO  FSNamesystem:906 - isPermissionEnabled    = fals
e
2026-09-13 23:47:20 INFO  FSNamesystem:907 - isStoragePolicyEnabled = true
2026-09-13 23:47:20 INFO  FSNamesystem:918 - HA Enabled: false
2026-09-13 23:47:20 INFO  Util:428 - dfs.datanode.fileio.profiling.samplin
g.percentage set to 0. Disabling file IO profiling
2026-09-13 23:47:20 INFO  DatanodeManager:412 - Slow peers collection thre
ad shutdown
2026-09-13 23:47:20 INFO  DatanodeManager:2182 - dfs.block.invalidate.limi
t : configured=1000, counted=60, effected=1000
2026-09-13 23:47:20 INFO  DatanodeManager:320 - dfs.namenode.datanode.regi
stration.ip-hostname-check=true
2026-09-13 23:47:20 INFO  BlockManager:77 - dfs.namenode.startup.delay.blo
ck.deletion.sec is set to 000:00:00:00.000
2026-09-13 23:47:20 INFO  BlockManager:83 - The block deletion will start
around 2026 Sep 13 23:47:20
2026-09-13 23:47:20 INFO  GSet:408 - Computing capacity for map BlocksMap
2026-09-13 23:47:20 INFO  GSet:409 - VM type       = 64-bit
2026-09-13 23:47:20 INFO  GSet:410 - 2.0% max memory 4 GB = 81.9 MB
2026-09-13 23:47:20 INFO  GSet:415 - capacity      = 2^23 = 8388608 entrie
s
2026-09-13 23:47:20 INFO  BlockManager:5778 - Storage policy satisfier is
disabled
2026-09-13 23:47:20 INFO  BlockManager:696 - dfs.block.access.token.enable
= false
2026-09-13 23:47:20 INFO  BlockManagerSafeMode:656 - Using 1000 as SafeMod
eMonitor Interval
2026-09-13 23:47:20 INFO  BlockManagerSafeMode:161 - dfs.namenode.safemod
e.threshold-pct = 0.999
2026-09-13 23:47:20 INFO  BlockManagerSafeMode:162 - dfs.namenode.safemod
e.min.datanodes = 0
2026-09-13 23:47:20 INFO  BlockManagerSafeMode:164 - dfs.namenode.safemod
e.extension = 30000
2026-09-13 23:47:20 INFO  BlockManager:682 - defaultReplication         =
1
2026-09-13 23:47:20 INFO  BlockManager:683 - maxReplication             =
512
2026-09-13 23:47:20 INFO  BlockManager:684 - minReplication             =
1
2026-09-13 23:47:20 INFO  BlockManager:685 - maxReplicationStreams      =
2
2026-09-13 23:47:20 INFO  BlockManager:686 - redundancyRecheckInterval  =
3000ms
2026-09-13 23:47:20 INFO  BlockManager:687 - encryptDataTransfer        =
false
2026-09-13 23:47:20 INFO  BlockManager:688 - maxNumBlocksToLog          =
1000
2026-09-13 23:47:20 INFO  FSDirectory:51 - GLOBAL serial map: bits=29 maxE
ntries=536870911
2026-09-13 23:47:20 INFO  FSDirectory:51 - USER serial map: bits=24 maxEnt
ries=16777215
2026-09-13 23:47:20 INFO  FSDirectory:51 - GROUP serial map: bits=24 maxEn
tries=16777215
2026-09-13 23:47:20 INFO  FSDirectory:51 - XATTR serial map: bits=24 maxEn
tries=16777215
```

```
2026-09-13 23:47:20 INFO  GSet:408 - Computing capacity for map INodeMap
2026-09-13 23:47:20 INFO  GSet:409 - VM type       = 64-bit
2026-09-13 23:47:20 INFO  GSet:410 - 1.0% max memory 4 GB = 41.0 MB
2026-09-13 23:47:20 INFO  GSet:415 - capacity      = 2^22 = 4194304 entrie
s
2026-09-13 23:47:20 INFO  FSDirectory:339 - ACLs enabled? true
2026-09-13 23:47:20 INFO  FSDirectory:343 - POSIX ACL inheritance enabled?
true
2026-09-13 23:47:20 INFO  FSDirectory:347 - XAttrs enabled? true
2026-09-13 23:47:20 INFO  NameNode:414 - Caching file names occurring more
than 10 times
2026-09-13 23:47:20 INFO  SnapshotManager:163 - Loaded config captureOpenF
iles: false, skipCaptureAccessTimeOnlyChange: false, snapshotDiffAllowSnap
RootDescendant: true, maxSnapshotFSLimit: 65536, maxSnapshotLimit: 65536
2026-09-13 23:47:20 INFO  SnapshotManager:176 - dfs.namenode.snapshot.dele
tion.ordered = false
2026-09-13 23:47:20 INFO  SnapshotManager:43 - SkipList is disabled
2026-09-13 23:47:20 INFO  GSet:408 - Computing capacity for map cachedBloc
ks
2026-09-13 23:47:20 INFO  GSet:409 - VM type       = 64-bit
2026-09-13 23:47:20 INFO  GSet:410 - 0.25% max memory 4 GB = 10.2 MB
2026-09-13 23:47:20 INFO  GSet:415 - capacity      = 2^20 = 1048576 entrie
s
2026-09-13 23:47:20 INFO  TopMetrics:76 - NNTop conf: dfs.namenode.top.win
dow.num.buckets = 10
2026-09-13 23:47:20 INFO  TopMetrics:78 - NNTop conf: dfs.namenode.top.nu
m.users = 10
2026-09-13 23:47:20 INFO  TopMetrics:80 - NNTop conf: dfs.namenode.top.win
dows.minutes = 1,5,25
2026-09-13 23:47:20 INFO  FSNamesystem:1158 - Retry cache on namenode is e
nabled
2026-09-13 23:47:20 INFO  FSNamesystem:1166 - Retry cache will use 0.03 of
total heap and retry cache entry expiry time is 600000 millis
2026-09-13 23:47:20 INFO  GSet:408 - Computing capacity for map NameNodeRe
tryCache
2026-09-13 23:47:20 INFO  GSet:409 - VM type       = 64-bit
2026-09-13 23:47:20 INFO  GSet:410 - 0.029999999329447746% max memory 4 GB
= 1.2 MB
2026-09-13 23:47:20 INFO  GSet:415 - capacity      = 2^17 = 131072 entries
2026-09-13 23:47:20 INFO  FSImage:186 - Allocated new BlockPoolId: BP-1211
785001-127.0.0.1-1789323440552
2026-09-13 23:47:20 INFO  Storage:595 - Storage directory /tmp/exp5-hdfs/n
amenode has been successfully formatted.
2026-09-13 23:47:20 INFO  FSImageFormatProtobuf:732 - Saving image file /t
mp/exp5-hdfs/namenode/current/fsimage.ckpt_0000000000000000000 using no co
mpression
2026-09-13 23:47:20 INFO  FSImageFormatProtobuf:736 - Image file /tmp/exp5
-hdfs/namenode/current/fsimage.ckpt_0000000000000000000 of size 399 bytes
saved in 0 seconds .
2026-09-13 23:47:20 INFO  NNStorageRetentionManager:205 - Going to retain
1 images with txid >= 0
2026-09-13 23:47:20 INFO  DatanodeManager:412 - Slow peers collection thre
ad shutdown
2026-09-13 23:47:20 INFO  FSNamesystem:1509 - Stopping services started fo
r active state
2026-09-13 23:47:20 INFO  FSNamesystem:1613 - Stopping services started fo
r standby state
2026-09-13 23:47:20 INFO  FSImage:1057 - FSImageSaver clean checkpoint: tx
id=0 when meet shutdown.
2026-09-13 23:47:20 INFO  NameNode:822 - SHUTDOWN_MSG:
```

```
/************************************************************
SHUTDOWN_MSG: Shutting down NameNode at Ojass-MacBook-Air.local/127.0.0.1
************************************************************/
```

Out[5]: CompletedProcess(args='hdfs namenode -format -force -nonInteractive', returncode=0, stdout="2026-09-13 23:47:20 INFO NameNode:809 - STARTUP_MSG: \n/**\nSTARTUP_MSG: Starting NameNode\nSTARTUP_MSG: host = Ojass-MacBook-Air.local/127.0.0.1\nSTARTUP_MSG: args = [-format, -force, -nonInteractive]\nSTARTUP_MSG: version = 3.4.1\nSTARTUP_MSG: classpath = /tmp/exp5-hadoop-conf:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jaxb-impl-2.2.3-1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-webapp-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/curator-client-5.2.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-handler-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/kerby-pkix-2.0.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-logging-1.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/reload4j-1.2.22.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jackson-core-2.12.7.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-all-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-redis-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jersey-json-1.22.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/slf4j-api-1.7.36.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-dns-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/dnsjava-3.6.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-util-ajax-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-stomp-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-compress-1.26.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/failureaccess-1.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/checker-qual-2.5.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jersey-servlet-1.19.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-native-epoll-4.1.100.Final-linux-aarch_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/kerby-asn1-2.0.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-cli-1.5.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/woodstox-core-5.4.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/zookeeper-jute-3.8.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/curator-framework-5.2.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/httpcore-4.4.13.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/kerby-config-2.0.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-smtp-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/kerb-util-2.0.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/hadoop-shaded-guava-1.3.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-mqtt-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/zookeeper-3.8.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-net-3.9.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/hadoop-annotations-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/javax.servlet-api-3.1.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jersey-server-1.19.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-io-2.16.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/hadoop-shaded-protobuf_3_25-1.3.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-server-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-util-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-rxtx-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/guava-27.0-jre.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-http-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-buffer-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/gson-2.9.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jcip-annotations-1.0-1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-common-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-io-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/c

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ommon/lib/jsr311-api-1.1.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/com
mon/lib/stax2-api-4.2.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/commo
n/lib/netty-codec-haproxy-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/sha
re/hadoop/common/lib/netty-codec-socks-4.1.100.Final.jar:/Users/ojas/had
oop-3.4.1/share/hadoop/common/lib/netty-transport-classes-epoll-4.1.100.
Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/hadoop-auth-
3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/nimbus-jose-j
wt-9.37.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-cod
ec-xml-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/li
b/netty-handler-ssl-ocsp-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/common/lib/netty-handler-proxy-4.1.100.Final.jar:/Users/ojas/ha
doop-3.4.1/share/hadoop/common/lib/netty-transport-4.1.100.Final.jar:/Us
ers/ojas/hadoop-3.4.1/share/hadoop/common/lib/jaxb-api-2.2.11.jar:/User
s/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-resolver-dns-4.1.100.F
inal.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jakarta.activa
tion-api-1.2.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/snap
py-java-1.1.10.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/ne
tty-resolver-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/com
mon/lib/listenablefuture-9999.0-empty-to-avoid-conflict-with-guava.jar:/
Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/metrics-core-3.2.4.jar:/
Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jsp-api-2.1.jar:/Users/o
jas/hadoop-3.4.1/share/hadoop/common/lib/avro-1.9.2.jar:/Users/ojas/hado
op-3.4.1/share/hadoop/common/lib/kerb-core-2.0.3.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/common/lib/jersey-core-1.19.4.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/common/lib/commons-math3-3.6.1.jar:/Users/ojas/hadoop
-3.4.1/share/hadoop/common/lib/netty-transport-native-kqueue-4.1.100.Fin
al-osx-aarch_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/net
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jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-nat
ive-unix-common-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/
common/lib/j2objc-annotations-1.1.jar:/Users/ojas/hadoop-3.4.1/share/had
oop/common/lib/jul-to-slf4j-1.7.36.jar:/Users/ojas/hadoop-3.4.1/share/ha
doop/common/lib/netty-transport-classes-kqueue-4.1.100.Final.jar:/Users/
ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-servlet-9.4.53.v2023100
9.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-codec-4.1.1
00.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-co
llections-3.2.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/ker
by-util-2.0.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commo
ns-codec-1.15.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetti
son-1.5.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/bcprov-jd
k18on-1.78.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/audien
ce-annotations-0.12.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/l
ib/netty-resolver-dns-native-macos-4.1.100.Final-osx-x86_64.jar:/Users/o
jas/hadoop-3.4.1/share/hadoop/common/lib/curator-recipes-5.2.0.jar:/User
s/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-resolver-dns-classes-m
acos-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/
commons-beanutils-1.9.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/commo
n/lib/netty-transport-native-epoll-4.1.100.Final-linux-x86_64.jar:/User
s/ojas/hadoop-3.4.1/share/hadoop/common/lib/kerb-crypto-2.0.3.jar:/User
s/ojas/hadoop-3.4.1/share/hadoop/common/lib/jackson-annotations-2.12.7.j
ar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jsch-0.1.55.jar:/Use
rs/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-udt-4.1.10
0.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jetty-secur
ity-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/li
b/jackson-databind-2.12.7.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/co
mmon/lib/netty-codec-memcache-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/common/lib/re2j-1.1.jar:/Users/ojas/hadoop-3.4.1/share/ha
doop/common/lib/commons-configuration2-2.10.1.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/common/lib/commons-lang3-3.12.0.jar:/Users/ojas/hadoop-
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3.4.1/share/hadoop/common/lib/httpclient-4.5.13.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/common/lib/animal-sniffer-annotations-1.17.jar:/User
s/ojas/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-native-epoll
-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/jett
y-http-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/commo
n/lib/jsr305-3.0.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/
netty-codec-http2-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoo
p/common/lib/commons-daemon-1.0.13.jar:/Users/ojas/hadoop-3.4.1/share/ha
doop/common/lib/jetty-xml-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/
share/hadoop/common/lib/netty-transport-native-kqueue-4.1.100.Final-osx-
x86_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/lib/commons-text
-1.10.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/hadoop-nfs-3.4.
1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/hadoop-kms-3.4.1.ja
r:/Users/ojas/hadoop-3.4.1/share/hadoop/common/hadoop-registry-3.4.1.ja
r:/Users/ojas/hadoop-3.4.1/share/hadoop/common/hadoop-common-3.4.1-test
s.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/common/hadoop-common-3.4.1.j
ar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs:/Users/ojas/hadoop-3.4.1/s
hare/hadoop/hdfs/lib/jaxb-impl-2.2.3-1.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/hdfs/lib/jetty-webapp-9.4.53.v20231009.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/hdfs/lib/curator-client-5.2.0.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/hdfs/lib/netty-handler-4.1.100.Final.jar:/Users/ojas/
hadoop-3.4.1/share/hadoop/hdfs/lib/kerby-pkix-2.0.3.jar:/Users/ojas/hado
op-3.4.1/share/hadoop/hdfs/lib/commons-logging-1.2.jar:/Users/ojas/hadoo
p-3.4.1/share/hadoop/hdfs/lib/reload4j-1.2.22.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/hdfs/lib/jackson-core-2.12.7.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/hdfs/lib/netty-all-4.1.100.Final.jar:/Users/ojas/hadoop
-3.4.1/share/hadoop/hdfs/lib/netty-codec-redis-4.1.100.Final.jar:/Users/
ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jersey-json-1.22.0.jar:/Users/oj
as/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-codec-dns-4.1.100.Final.ja
r:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/dnsjava-3.6.1.jar:/User
s/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jetty-util-ajax-9.4.53.v202310
09.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-codec-stomp-
4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/commons
-compress-1.26.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/fail
ureaccess-1.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/checker
-qual-2.5.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jersey-se
rvlet-1.19.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-tr
ansport-native-epoll-4.1.100.Final-linux-aarch_64.jar:/Users/ojas/hadoop
-3.4.1/share/hadoop/hdfs/lib/kerby-asn1-2.0.3.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/hdfs/lib/commons-cli-1.5.0.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/hdfs/lib/woodstox-core-5.4.0.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/hdfs/lib/zookeeper-jute-3.8.4.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/hdfs/lib/curator-framework-5.2.0.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/hdfs/lib/httpcore-4.4.13.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/hdfs/lib/kerby-config-2.0.3.jar:/Users/ojas/hadoop-3.4.1/
share/hadoop/hdfs/lib/netty-codec-smtp-4.1.100.Final.jar:/Users/ojas/had
oop-3.4.1/share/hadoop/hdfs/lib/kerb-util-2.0.3.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/hdfs/lib/hadoop-shaded-guava-1.3.0.jar:/Users/ojas/ha
doop-3.4.1/share/hadoop/hdfs/lib/netty-codec-mqtt-4.1.100.Final.jar:/Use
rs/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/zookeeper-3.8.4.jar:/Users/oj
as/hadoop-3.4.1/share/hadoop/hdfs/lib/commons-net-3.9.0.jar:/Users/ojas/
hadoop-3.4.1/share/hadoop/hdfs/lib/hadoop-annotations-3.4.1.jar:/Users/o
jas/hadoop-3.4.1/share/hadoop/hdfs/lib/javax.servlet-api-3.1.0.jar:/User
s/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jersey-server-1.19.4.jar:/User
s/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/commons-io-2.16.1.jar:/Users/o
jas/hadoop-3.4.1/share/hadoop/hdfs/lib/hadoop-shaded-protobuf_3_25-1.3.
0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jetty-server-9.4.5
3.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jetty-uti
l-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/ne
tty-transport-rxtx-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hado
```

```
op/hdfs/lib/guava-27.0-jre.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdf
s/lib/netty-codec-http-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/
hadoop/hdfs/lib/netty-buffer-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/
share/hadoop/hdfs/lib/gson-2.9.0.jar:/Users/ojas/hadoop-3.4.1/share/hado
op/hdfs/lib/jcip-annotations-1.0-1.jar:/Users/ojas/hadoop-3.4.1/share/ha
doop/hdfs/lib/netty-common-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/sh
are/hadoop/hdfs/lib/jetty-io-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/hdfs/lib/jsr311-api-1.1.1.jar:/Users/ojas/hadoop-3.4.1/
share/hadoop/hdfs/lib/stax2-api-4.2.1.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/hdfs/lib/netty-codec-haproxy-4.1.100.Final.jar:/Users/ojas/hado
op-3.4.1/share/hadoop/hdfs/lib/netty-codec-socks-4.1.100.Final.jar:/User
s/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-transport-classes-epoll-
4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/hadoop-
auth-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/nimbus-jos
e-jwt-9.37.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-co
dec-xml-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/li
b/netty-handler-ssl-ocsp-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/hdfs/lib/netty-handler-proxy-4.1.100.Final.jar:/Users/ojas/hado
op-3.4.1/share/hadoop/hdfs/lib/netty-transport-4.1.100.Final.jar:/Users/
ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jaxb-api-2.2.11.jar:/Users/ojas/
hadoop-3.4.1/share/hadoop/hdfs/lib/json-simple-1.1.1.jar:/Users/ojas/had
oop-3.4.1/share/hadoop/hdfs/lib/netty-resolver-dns-4.1.100.Final.jar:/Us
ers/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jakarta.activation-api-1.2.
1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/snappy-java-1.1.10.
4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-resolver-4.1.
100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/listenablef
uture-9999.0-empty-to-avoid-conflict-with-guava.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/hdfs/lib/metrics-core-3.2.4.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/hdfs/lib/avro-1.9.2.jar:/Users/ojas/hadoop-3.4.1/share/
hadoop/hdfs/lib/kerb-core-2.0.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoo
p/hdfs/lib/jersey-core-1.19.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/
hdfs/lib/commons-math3-3.6.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/h
dfs/lib/netty-transport-native-kqueue-4.1.100.Final-osx-aarch_64.jar:/Us
ers/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-resolver-dns-native-ma
cos-4.1.100.Final-osx-aarch_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoo
p/hdfs/lib/netty-transport-sctp-4.1.100.Final.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/hdfs/lib/netty-transport-native-unix-common-4.1.100.Fin
al.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/j2objc-annotations
-1.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-transport-
classes-kqueue-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/h
dfs/lib/jetty-servlet-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/hdfs/lib/netty-codec-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/hdfs/lib/commons-collections-3.2.2.jar:/Users/ojas/hadoop
-3.4.1/share/hadoop/hdfs/lib/kerby-util-2.0.3.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/hdfs/lib/commons-codec-1.15.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/hdfs/lib/jettison-1.5.4.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/hdfs/lib/audience-annotations-0.12.0.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/hdfs/lib/netty-resolver-dns-native-macos-4.1.100.Final-
osx-x86_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/curator-re
cipes-5.2.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-res
olver-dns-classes-macos-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/hdfs/lib/commons-beanutils-1.9.4.jar:/Users/ojas/hadoop-3.4.1/s
hare/hadoop/hdfs/lib/netty-transport-native-epoll-4.1.100.Final-linux-x8
6_64.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/kerb-crypto-2.0.
3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jackson-annotations
-2.12.7.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jsch-0.1.55.j
ar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-transport-udt-4.
1.100.Final.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jetty-sec
urity-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/li
b/jackson-databind-2.12.7.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hd
```

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fs/lib/netty-codec-memcache-4.1.100.Final.jar:/Users/ojas/hadoop-3.4.1/s
hare/hadoop/hdfs/lib/re2j-1.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/
hdfs/lib/commons-configuration2-2.10.1.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/hdfs/lib/commons-lang3-3.12.0.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/hdfs/lib/httpclient-4.5.13.jar:/Users/ojas/hadoop-3.4.1/share/h
adoop/hdfs/lib/animal-sniffer-annotations-1.17.jar:/Users/ojas/hadoop-3.
4.1/share/hadoop/hdfs/lib/leveldbjni-all-1.8.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/hdfs/lib/netty-transport-native-epoll-4.1.100.Final.jar:/
Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jetty-http-9.4.53.v2023100
9.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jsr305-3.0.2.jar:/U
sers/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/HikariCP-4.0.3.jar:/Users/o
jas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-codec-http2-4.1.100.Final.j
ar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/commons-daemon-1.0.13.
jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/jetty-xml-9.4.53.v202
31009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/hdfs/lib/netty-transport
-native-kqueue-4.1.100.Final-osx-x86_64.jar:/Users/ojas/hadoop-3.4.1/sha
re/hadoop/hdfs/lib/commons-text-1.10.0.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/hdfs/hadoop-hdfs-native-client-3.4.1-tests.jar:/Users/ojas/hado
op-3.4.1/share/hadoop/hdfs/hadoop-hdfs-rbf-3.4.1.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/hdfs/hadoop-hdfs-native-client-3.4.1.jar:/Users/ojas/
hadoop-3.4.1/share/hadoop/hdfs/hadoop-hdfs-client-3.4.1.jar:/Users/ojas/
hadoop-3.4.1/share/hadoop/hdfs/hadoop-hdfs-rbf-3.4.1-tests.jar:/Users/oj
as/hadoop-3.4.1/share/hadoop/hdfs/hadoop-hdfs-3.4.1.jar:/Users/ojas/hado
op-3.4.1/share/hadoop/hdfs/hadoop-hdfs-client-3.4.1-tests.jar:/Users/oja
s/hadoop-3.4.1/share/hadoop/hdfs/hadoop-hdfs-3.4.1-tests.jar:/Users/oja
s/hadoop-3.4.1/share/hadoop/hdfs/hadoop-hdfs-nfs-3.4.1.jar:/Users/ojas/h
adoop-3.4.1/share/hadoop/hdfs/hadoop-hdfs-httpfs-3.4.1.jar:/Users/ojas/h
adoop-3.4.1/share/hadoop/mapreduce/hadoop-mapreduce-client-shuffle-3.4.
1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/mapreduce/hadoop-mapreduce-c
lient-common-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/mapreduce/h
adoop-mapreduce-client-jobclient-3.4.1-tests.jar:/Users/ojas/hadoop-3.4.
1/share/hadoop/mapreduce/hadoop-mapreduce-client-jobclient-3.4.1.jar:/Us
ers/ojas/hadoop-3.4.1/share/hadoop/mapreduce/hadoop-mapreduce-client-nat
ivetask-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/mapreduce/hadoop
-mapreduce-client-app-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/ma
preduce/hadoop-mapreduce-client-hs-plugins-3.4.1.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/mapreduce/hadoop-mapreduce-client-hs-3.4.1.jar:/User
s/ojas/hadoop-3.4.1/share/hadoop/mapreduce/hadoop-mapreduce-client-uploa
der-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/mapreduce/hadoop-map
reduce-examples-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/mapreduc
e/hadoop-mapreduce-client-core-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/
hadoop/yarn:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/javax.websock
et-api-1.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/guice-serv
let-4.2.3.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jakarta.xm
l.bind-api-2.3.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jack
son-jaxrs-base-2.12.7.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/li
b/FastInfoset-1.2.15.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/
websocket-servlet-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/ha
doop/yarn/lib/aopalliance-1.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/
yarn/lib/websocket-api-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/sha
re/hadoop/yarn/lib/jaxb-runtime-2.3.1.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/yarn/lib/bcpkix-jdk18on-1.78.1.jar:/Users/ojas/hadoop-3.4.1/sha
re/hadoop/yarn/lib/istack-commons-runtime-3.0.7.jar:/Users/ojas/hadoop-
3.4.1/share/hadoop/yarn/lib/snakeyaml-2.0.jar:/Users/ojas/hadoop-3.4.1/s
hare/hadoop/yarn/lib/guice-4.2.3.jar:/Users/ojas/hadoop-3.4.1/share/hado
op/yarn/lib/javax.activation-api-1.2.0.jar:/Users/ojas/hadoop-3.4.1/shar
e/hadoop/yarn/lib/codemodel-2.6.jar:/Users/ojas/hadoop-3.4.1/share/hadoo
p/yarn/lib/asm-commons-9.6.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yar
n/lib/jackson-jaxrs-json-provider-2.12.7.jar:/Users/ojas/hadoop-3.4.1/sh
are/hadoop/yarn/lib/objenesis-2.6.jar:/Users/ojas/hadoop-3.4.1/share/had
```

oop/yarn/lib/jline-3.9.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/ehcache-3.8.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/websocket-common-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/asm-tree-9.6.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/fst-2.50.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jetty-annotations-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/mssql-jdbc-6.2.1.jre7.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/websocket-client-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jsonschema2pojo-core-1.0.2.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/stax-ex-1.8.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jetty-plus-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jersey-client-1.19.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/javax.websocket-client-api-1.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/bcutil-jdk18on-1.78.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jetty-jndi-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/javax-websocket-server-impl-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/commons-lang-2.6.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jna-5.2.0.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/cache-api-1.1.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/websocket-server-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/javax-websocket-client-impl-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jersey-guice-1.19.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jackson-module-jaxb-annotations-2.12.7.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/javax.inject-1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/jetty-client-9.4.53.v20231009.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/swagger-annotations-1.5.4.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/lib/txw2-2.3.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-common-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-web-proxy-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-nodemanager-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-resourcemanager-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-common-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-router-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-applications-mawo-core-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-services-core-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-client-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-services-api-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-timeline-pluginstorage-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-applications-distributedshell-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-globalpolicygenerator-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-sharedcachemanager-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-registry-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-applicationhistoryservice-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-applications-unmanaged-am-launcher-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-server-tests-3.4.1.jar:/Users/ojas/hadoop-3.4.1/share/hadoop/yarn/hadoop-yarn-api-3.4.1.jar\nSTARTUP_MSG: build = https://github.com/apache/hadoop.git -r 4d7825309348956336b8f06a08322b78422849b1; compiled by 'mthakur' on 2024-10-09T14:57Z\nSTARTUP_MSG: java = 17.0.19\n**/\n2026-09-13 23:47:20 INFO NameNode:91 - registered UNIX signal handlers for [TERM, HUP, INT]\n2026-09-13 23:47:20 INFO NameNode:1846 - createNameNode [-format, -force, -nonInteractive]\n2026-09-13 23:47:20 WARN NativeCodeLoader:60 - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable\n2026-09-13 23:47:20 INFO NameNode:1393 - Formatting using clusterid: CID-30f97f04-b1e6-4a5c-846c-7e0c79163e13\n2026-09-13 23:47:20 INFO F

SEditLog:238 - Edit logging is async:true\n2026-09-13 23:47:20 INFO FSN
amesystem:869 - KeyProvider: null\n2026-09-13 23:47:20 INFO FSNamesyste
m:142 - fsLock is fair: true\n2026-09-13 23:47:20 INFO FSNamesystem:160
- Detailed lock hold time metrics enabled: false\n2026-09-13 23:47:20 IN
FO FSNamesystem:904 - fsOwner   = ojas (auth:SIMPLE)\n2026
-09-13 23:47:20 INFO FSNamesystem:905 - supergroup   = superg
roup\n2026-09-13 23:47:20 INFO FSNamesystem:906 - isPermissionEnabled
= false\n2026-09-13 23:47:20 INFO FSNamesystem:907 - isStoragePolicyEna
bled = true\n2026-09-13 23:47:20 INFO FSNamesystem:918 - HA Enabled: fa
lse\n2026-09-13 23:47:20 INFO Util:428 - dfs.datanode.fileio.profiling.
sampling.percentage set to 0. Disabling file IO profiling\n2026-09-13 2
3:47:20 INFO DatanodeManager:412 - Slow peers collection thread shutdow
n\n2026-09-13 23:47:20 INFO DatanodeManager:2182 - dfs.block.invalidat
e.limit : configured=1000, counted=60, effected=1000\n2026-09-13 23:47:2
0 INFO DatanodeManager:320 - dfs.namenode.datanode.registration.ip-host
name-check=true\n2026-09-13 23:47:20 INFO BlockManager:77 - dfs.namenod
e.startup.delay.block.deletion.sec is set to 000:00:00:00.000\n2026-09-1
3 23:47:20 INFO BlockManager:83 - The block deletion will start around
2026 Sep 13 23:47:20\n2026-09-13 23:47:20 INFO GSet:408 - Computing cap
acity for map BlocksMap\n2026-09-13 23:47:20 INFO GSet:409 - VM type
= 64-bit\n2026-09-13 23:47:20 INFO GSet:410 - 2.0% max memory 4 GB = 8
1.9 MB\n2026-09-13 23:47:20 INFO GSet:415 - capacity   = 2^23 = 8388
608 entries\n2026-09-13 23:47:20 INFO BlockManager:5778 - Storage polic
y satisfier is disabled\n2026-09-13 23:47:20 INFO BlockManager:696 - df
s.block.access.token.enable = false\n2026-09-13 23:47:20 INFO BlockMana
gerSafeMode:656 - Using 1000 as SafeModeMonitor Interval\n2026-09-13 23:
47:20 INFO BlockManagerSafeMode:161 - dfs.namenode.safemode.threshold-p
ct = 0.999\n2026-09-13 23:47:20 INFO BlockManagerSafeMode:162 - dfs.nam
enode.safemode.min.datanodes = 0\n2026-09-13 23:47:20 INFO BlockManager
SafeMode:164 - dfs.namenode.safemode.extension = 30000\n2026-09-13 23:4
7:20 INFO BlockManager:682 - defaultReplication   = 1\n2026-09-13
23:47:20 INFO BlockManager:683 - maxReplication   = 512\n2026
-09-13 23:47:20 INFO BlockManager:684 - minReplication   = 1
\n2026-09-13 23:47:20 INFO BlockManager:685 - maxReplicationStreams
= 2\n2026-09-13 23:47:20 INFO BlockManager:686 - redundancyRecheckInter
val = 3000ms\n2026-09-13 23:47:20 INFO BlockManager:687 - encryptDataT
ransfer   = false\n2026-09-13 23:47:20 INFO BlockManager:688 - max
NumBlocksToLog   = 1000\n2026-09-13 23:47:20 INFO FSDirectory:51
- GLOBAL serial map: bits=29 maxEntries=536870911\n2026-09-13 23:47:20 I
NFO FSDirectory:51 - USER serial map: bits=24 maxEntries=16777215\n2026
-09-13 23:47:20 INFO FSDirectory:51 - GROUP serial map: bits=24 maxEntr
ies=16777215\n2026-09-13 23:47:20 INFO FSDirectory:51 - XATTR serial ma
p: bits=24 maxEntries=16777215\n2026-09-13 23:47:20 INFO GSet:408 - Com
puting capacity for map INodeMap\n2026-09-13 23:47:20 INFO GSet:409 - V
M type   = 64-bit\n2026-09-13 23:47:20 INFO GSet:410 - 1.0% max mem
ory 4 GB = 41.0 MB\n2026-09-13 23:47:20 INFO GSet:415 - capacity   =
2^22 = 4194304 entries\n2026-09-13 23:47:20 INFO FSDirectory:339 - ACLs
enabled? true\n2026-09-13 23:47:20 INFO FSDirectory:343 - POSIX ACL inh
eritance enabled? true\n2026-09-13 23:47:20 INFO FSDirectory:347 - XAtt
rs enabled? true\n2026-09-13 23:47:20 INFO NameNode:414 - Caching file
names occurring more than 10 times\n2026-09-13 23:47:20 INFO SnapshotMa
nager:163 - Loaded config captureOpenFiles: false, skipCaptureAccessTime
OnlyChange: false, snapshotDiffAllowSnapRootDescendant: true, maxSnapsho
tFSLimit: 65536, maxSnapshotLimit: 65536\n2026-09-13 23:47:20 INFO Snap
shotManager:176 - dfs.namenode.snapshot.deletion.ordered = false\n2026-0
9-13 23:47:20 INFO SnapshotManager:43 - SkipList is disabled\n2026-09-1
3 23:47:20 INFO GSet:408 - Computing capacity for map cachedBlocks\n202
6-09-13 23:47:20 INFO GSet:409 - VM type   = 64-bit\n2026-09-13 23:
47:20 INFO GSet:410 - 0.25% max memory 4 GB = 10.2 MB\n2026-09-13 23:4
7:20 INFO GSet:415 - capacity   = 2^20 = 1048576 entries\n2026-09-13

```
23:47:20 INFO  TopMetrics:76 - NNTop conf: dfs.namenode.top.window.num.b
uckets = 10\n2026-09-13 23:47:20 INFO  TopMetrics:78 - NNTop conf: dfs.n
amenode.top.num.users = 10\n2026-09-13 23:47:20 INFO  TopMetrics:80 - NN
Top conf: dfs.namenode.top.windows.minutes = 1,5,25\n2026-09-13 23:47:20
INFO  FSNamesystem:1158 - Retry cache on namenode is enabled\n2026-09-13
23:47:20 INFO  FSNamesystem:1166 - Retry cache will use 0.03 of total he
ap and retry cache entry expiry time is 600000 millis\n2026-09-13 23:47:
20 INFO  GSet:408 - Computing capacity for map NameNodeRetryCache\n2026-
09-13 23:47:20 INFO  GSet:409 - VM type       = 64-bit\n2026-09-13 23:4
7:20 INFO  GSet:410 - 0.029999999329447746% max memory 4 GB = 1.2 MB\n20
26-09-13 23:47:20 INFO  GSet:415 - capacity      = 2^17 = 131072 entries
\n2026-09-13 23:47:20 INFO  FSImage:186 - Allocated new BlockPoolId: BP-
1211785001-127.0.0.1-1789323440552\n2026-09-13 23:47:20 INFO  Storage:59
5 - Storage directory /tmp/exp5-hdfs/namenode has been successfully form
atted.\n2026-09-13 23:47:20 INFO  FSImageFormatProtobuf:732 - Saving ima
ge file /tmp/exp5-hdfs/namenode/current/fsimage.ckpt_0000000000000000000
using no compression\n2026-09-13 23:47:20 INFO  FSImageFormatProtobuf:73
6 - Image file /tmp/exp5-hdfs/namenode/current/fsimage.ckpt_000000000000
0000000 of size 399 bytes saved in 0 seconds .\n2026-09-13 23:47:20 INFO
NNStorageRetentionManager:205 - Going to retain 1 images with txid >= 0
\n2026-09-13 23:47:20 INFO  DatanodeManager:412 - Slow peers collection
thread shutdown\n2026-09-13 23:47:20 INFO  FSNamesystem:1509 - Stopping
services started for active state\n2026-09-13 23:47:20 INFO  FSNamesyste
m:1613 - Stopping services started for standby state\n2026-09-13 23:47:2
0 INFO  FSImage:1057 - FSImageSaver clean checkpoint: txid=0 when meet s
hutdown.\n2026-09-13 23:47:20 INFO  NameNode:822 - SHUTDOWN_MSG: \n/****
************************************************************\nSHUTDOWN_MSG:
Shutting down NameNode at Ojass-MacBook-Air.local/127.0.0.1\n************
************************************************/\n", stderr='')
```

7. Start Hadoop HDFS Services

In [6]:

```
run("hdfs --daemon start namenode")
run("hdfs --daemon start datanode")

for _ in range(20):
    time.sleep(1)
    report = subprocess.run(
        "hdfs dfsadmin -report",
        shell=True,
        text=True,
        capture_output=True,
        env=os.environ.copy(),
    )
    if report.returncode == 0 and "Live datanodes" in (report.stdout + re
        break

print("HDFS daemons started.\n")
run("jps")
```

```
$ hdfs --daemon start namenode
$ hdfs --daemon start datanode
HDFS daemons started.

$ jps
57451 DataNode
57501 Jps
57390 NameNode
```

Out[6]: CompletedProcess(args='jps', returncode=0, stdout='57451 DataNode\n57501 Jps\n57390 NameNode\n', stderr='')

8. Verify HDFS Is Running

In [7]:
```
run("hdfs dfsadmin -report")
```

```
$ hdfs dfsadmin -report
2026-09-13 23:47:27 WARN  NativeCodeLoader:60 - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Configured Capacity: 494384795648 (460.43 GB)
Present Capacity: 121624821760 (113.27 GB)
DFS Remaining: 121624817664 (113.27 GB)
DFS Used: 4096 (4 KB)
DFS Used%: 0.00%
Replicated Blocks:
        Under replicated blocks: 0
        Blocks with corrupt replicas: 0
        Missing blocks: 0
        Missing blocks (with replication factor 1): 0
        Low redundancy blocks with highest priority to recover: 0
        Pending deletion blocks: 0
Erasure Coded Block Groups:
        Low redundancy block groups: 0
        Block groups with corrupt internal blocks: 0
        Missing block groups: 0
        Low redundancy blocks with highest priority to recover: 0
        Pending deletion blocks: 0

-------------------------------------------------
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: localhost
Decommission Status : Normal
Configured Capacity: 494384795648 (460.43 GB)
DFS Used: 4096 (4 KB)
Non DFS Used: 372759973888 (347.16 GB)
DFS Remaining: 121624817664 (113.27 GB)
DFS Used%: 0.00%
DFS Remaining%: 24.60%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sun Sep 13 23:47:26 IST 2026
Last Block Report: Sun Sep 13 23:47:23 IST 2026
Num of Blocks: 0
```

Out[7]:
```
CompletedProcess(args='hdfs dfsadmin -report', returncode=0, stdout='202
6-09-13 23:47:27 WARN  NativeCodeLoader:60 - Unable to load native-hadoo
p library for your platform... using builtin-java classes where applicab
le\nConfigured Capacity: 494384795648 (460.43 GB)\nPresent Capacity: 121
624821760 (113.27 GB)\nDFS Remaining: 121624817664 (113.27 GB)\nDFS Use
d: 4096 (4 KB)\nDFS Used%: 0.00%\nReplicated Blocks:\n\tUnder replicated
blocks: 0\n\tBlocks with corrupt replicas: 0\n\tMissing blocks: 0\n\tMis
sing blocks (with replication factor 1): 0\n\tLow redundancy blocks with
highest priority to recover: 0\n\tPending deletion blocks: 0\nErasure Co
ded Block Groups: \n\tLow redundancy block groups: 0\n\tBlock groups wit
h corrupt internal blocks: 0\n\tMissing block groups: 0\n\tLow redundanc
y blocks with highest priority to recover: 0\n\tPending deletion blocks:
0\n\n-------------------------------------------------\nLive datanodes
(1):\n\nName: 127.0.0.1:9866 (localhost)\nHostname: localhost\nDecommiss
ion Status : Normal\nConfigured Capacity: 494384795648 (460.43 GB)\nDFS
Used: 4096 (4 KB)\nNon DFS Used: 372759973888 (347.16 GB)\nDFS Remainin
g: 121624817664 (113.27 GB)\nDFS Used%: 0.00%\nDFS Remaining%: 24.60%\nC
onfigured Cache Capacity: 0 (0 B)\nCache Used: 0 (0 B)\nCache Remaining:
0 (0 B)\nCache Used%: 100.00%\nCache Remaining%: 0.00%\nXceivers: 0\nLas
t contact: Sun Sep 13 23:47:26 IST 2026\nLast Block Report: Sun Sep 13 2
3:47:23 IST 2026\nNum of Blocks: 0\n\n\n', stderr='')
```

9. Create Directories in HDFS

In [8]:
```
run("hdfs dfs -mkdir -p /user/lab/input")
run("hdfs dfs -mkdir -p /user/lab/output")
run("hdfs dfs -ls -R /user")
```

```
$ hdfs dfs -mkdir -p /user/lab/input
2026-09-13 23:47:27 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
$ hdfs dfs -mkdir -p /user/lab/output
2026-09-13 23:47:28 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
$ hdfs dfs -ls -R /user
2026-09-13 23:47:28 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/input
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/outpu
t
```

Out[8]:
```
CompletedProcess(args='hdfs dfs -ls -R /user', returncode=0, stdout='202
6-09-13 23:47:28 WARN  NativeCodeLoader:60 - Unable to load native-hadoo
p library for your platform... using builtin-java classes where applicab
le\ndrwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab
\ndrwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/i
nput\ndrwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/l
ab/output\n', stderr='')
```

10. Upload Files to HDFS

In [9]:
```
run(f"hdfs dfs -put -f {quote(str(LOCAL_DIR / 'sample.txt'))} /user/lab/i
run("hdfs dfs -ls /user/lab/input")
```

```
$ hdfs dfs -put -f /tmp/exp5-local/sample.txt /user/lab/input/
2026-09-13 23:47:29 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
$ hdfs dfs -ls /user/lab/input
2026-09-13 23:47:30 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Found 1 items
-rw-r--r--   1 ojas supergroup        262 2026-09-13 23:47 /user/lab/inpu
t/sample.txt
```

Out[9]:
```
CompletedProcess(args='hdfs dfs -ls /user/lab/input', returncode=0, stdo
ut='2026-09-13 23:47:30 WARN  NativeCodeLoader:60 - Unable to load nativ
e-hadoop library for your platform... using builtin-java classes where a
pplicable\nFound 1 items\n-rw-r--r--   1 ojas supergroup        262 2026
-09-13 23:47 /user/lab/input/sample.txt\n', stderr='')
```

11. Display HDFS File Contents

In [10]:
```
run("hdfs dfs -cat /user/lab/input/sample.txt")
```

```
$ hdfs dfs -cat /user/lab/input/sample.txt
2026-09-13 23:47:30 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
This is a sample text file for Hadoop HDFS operations.
Experiment 5: Hadoop Installation, Configuration, and HDFS Operations

HDFS stores files by dividing them into blocks and distributing them acros
s nodes.
Common operations: mkdir, put, ls, cat, get, cp, rm.
```

Out[10]:
```
CompletedProcess(args='hdfs dfs -cat /user/lab/input/sample.txt', return
code=0, stdout='2026-09-13 23:47:30 WARN  NativeCodeLoader:60 - Unable t
o load native-hadoop library for your platform... using builtin-java cla
sses where applicable\nThis is a sample text file for Hadoop HDFS operat
ions.\nExperiment 5: Hadoop Installation, Configuration, and HDFS Operat
ions\n\nHDFS stores files by dividing them into blocks and distributing
them across nodes.\nCommon operations: mkdir, put, ls, cat, get, cp, r
m.\n', stderr='')
```

12. Copy, Download, and Delete Files

In [11]:
```
run("hdfs dfs -cp /user/lab/input/sample.txt /user/lab/output/sample_copy
print("\nCopied inside HDFS. Listing /user/lab/output:")
run("hdfs dfs -ls /user/lab/output")

print("\nDownloading to local system:")
local_copy = DOWNLOAD_DIR / "sample_from_hdfs.txt"
if local_copy.exists():
    local_copy.unlink()
run(f"hdfs dfs -get /user/lab/output/sample_copy.txt {quote(str(local_cop
print("Local file contents:")
print(local_copy.read_text())

print("Deleting the copy from HDFS:")
run("hdfs dfs -rm /user/lab/output/sample_copy.txt")
run("hdfs dfs -ls /user/lab/output")
```

```
$ hdfs dfs -cp /user/lab/input/sample.txt /user/lab/output/sample_copy.txt
2026-09-13 23:47:31 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e

Copied inside HDFS. Listing /user/lab/output:
$ hdfs dfs -ls /user/lab/output
2026-09-13 23:47:32 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Found 1 items
-rw-r--r--   1 ojas supergroup        262 2026-09-13 23:47 /user/lab/outpu
t/sample_copy.txt

Downloading to local system:
$ hdfs dfs -get /user/lab/output/sample_copy.txt /tmp/exp5-downloads/sampl
e_from_hdfs.txt
2026-09-13 23:47:32 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Local file contents:
This is a sample text file for Hadoop HDFS operations.
Experiment 5: Hadoop Installation, Configuration, and HDFS Operations

HDFS stores files by dividing them into blocks and distributing them acros
s nodes.
Common operations: mkdir, put, ls, cat, get, cp, rm.

Deleting the copy from HDFS:
$ hdfs dfs -rm /user/lab/output/sample_copy.txt
2026-09-13 23:47:33 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Deleted /user/lab/output/sample_copy.txt
$ hdfs dfs -ls /user/lab/output
2026-09-13 23:47:33 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
```

Out[11]:
```
CompletedProcess(args='hdfs dfs -ls /user/lab/output', returncode=0, std
out='2026-09-13 23:47:33 WARN  NativeCodeLoader:60 - Unable to load nati
ve-hadoop library for your platform... using builtin-java classes where
applicable\n', stderr='')
```

Exercises / Assignments

Exercise 1: Create a new directory in HDFS and upload multiple files

In [12]:
```
run("hdfs dfs -mkdir -p /user/lab/exercise1")
run(
    "hdfs dfs -put -f "
    f"{quote(str(LOCAL_DIR / 'file1.txt'))} {quote(str(LOCAL_DIR / 'file2
    "/user/lab/exercise1/"
)
run("hdfs dfs -ls /user/lab/exercise1")
```

```
$ hdfs dfs -mkdir -p /user/lab/exercise1
2026-09-13 23:47:34 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
$ hdfs dfs -put -f /tmp/exp5-local/file1.txt /tmp/exp5-local/file2.txt /us
er/lab/exercise1/
2026-09-13 23:47:34 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
$ hdfs dfs -ls /user/lab/exercise1
2026-09-13 23:47:35 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Found 2 items
-rw-r--r--   1 ojas supergroup        146 2026-09-13 23:47 /user/lab/exerc
ise1/file1.txt
-rw-r--r--   1 ojas supergroup        136 2026-09-13 23:47 /user/lab/exerc
ise1/file2.txt
```

Out[12]:
```
CompletedProcess(args='hdfs dfs -ls /user/lab/exercise1', returncode=0,
stdout='2026-09-13 23:47:35 WARN  NativeCodeLoader:60 - Unable to load n
ative-hadoop library for your platform... using builtin-java classes whe
re applicable\nFound 2 items\n-rw-r--r--   1 ojas supergroup        146
2026-09-13 23:47 /user/lab/exercise1/file1.txt\n-rw-r--r--   1 ojas supe
rgroup        136 2026-09-13 23:47 /user/lab/exercise1/file2.txt\n', std
err='')
```

Exercise 2: Copy a file from one HDFS directory to another

In [13]:
```
run("hdfs dfs -mkdir -p /user/lab/exercise2")
run("hdfs dfs -cp /user/lab/exercise1/file1.txt /user/lab/exercise2/file1
run("hdfs dfs -ls /user/lab/exercise2")
```

```
$ hdfs dfs -mkdir -p /user/lab/exercise2
2026-09-13 23:47:36 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
$ hdfs dfs -cp /user/lab/exercise1/file1.txt /user/lab/exercise2/file1_cop
ied.txt
2026-09-13 23:47:36 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
$ hdfs dfs -ls /user/lab/exercise2
2026-09-13 23:47:37 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Found 1 items
-rw-r--r--   1 ojas supergroup        146 2026-09-13 23:47 /user/lab/exerc
ise2/file1_copied.txt
```

Out[13]:
```
CompletedProcess(args='hdfs dfs -ls /user/lab/exercise2', returncode=0,
stdout='2026-09-13 23:47:37 WARN  NativeCodeLoader:60 - Unable to load n
ative-hadoop library for your platform... using builtin-java classes whe
re applicable\nFound 1 items\n-rw-r--r--   1 ojas supergroup        146
2026-09-13 23:47 /user/lab/exercise2/file1_copied.txt\n', stderr='')
```

Exercise 3: Delete a directory from HDFS

In [14]:
```
run("hdfs dfs -mkdir -p /user/lab/to_delete")
run(f"hdfs dfs -put -f {quote(str(LOCAL_DIR / 'sample.txt'))} /user/lab/t
```

```
print("Before delete:")
run("hdfs dfs -ls /user/lab")
run("hdfs dfs -rm -r /user/lab/to_delete")
print("\nAfter delete:")
run("hdfs dfs -ls /user/lab")
```

```
$ hdfs dfs -mkdir -p /user/lab/to_delete
2026-09-13 23:47:38 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
$ hdfs dfs -put -f /tmp/exp5-local/sample.txt /user/lab/to_delete/
2026-09-13 23:47:38 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Before delete:
$ hdfs dfs -ls /user/lab
2026-09-13 23:47:39 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Found 5 items
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/exerc
ise1
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/exerc
ise2
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/input
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/outpu
t
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/to_de
lete
$ hdfs dfs -rm -r /user/lab/to_delete
2026-09-13 23:47:39 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Deleted /user/lab/to_delete

After delete:
$ hdfs dfs -ls /user/lab
2026-09-13 23:47:40 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Found 4 items
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/exerc
ise1
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/exerc
ise2
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/input
drwxr-xr-x   - ojas supergroup          0 2026-09-13 23:47 /user/lab/outpu
t
```

Out[14]:
```
CompletedProcess(args='hdfs dfs -ls /user/lab', returncode=0, stdout='20
26-09-13 23:47:40 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applica
ble\nFound 4 items\ndrwxr-xr-x   - ojas supergroup          0 2026-09-13
23:47 /user/lab/exercise1\ndrwxr-xr-x   - ojas supergroup          0 202
6-09-13 23:47 /user/lab/exercise2\ndrwxr-xr-x   - ojas supergroup
0 2026-09-13 23:47 /user/lab/input\ndrwxr-xr-x   - ojas supergroup
0 2026-09-13 23:47 /user/lab/output\n', stderr='')
```

Exercise 4: Display the contents of an HDFS file

In [15]:
```
run("hdfs dfs -cat /user/lab/exercise1/file2.txt")
```

```
$ hdfs dfs -cat /user/lab/exercise1/file2.txt
2026-09-13 23:47:40 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
File 2: hadoop notes
HDFS = Hadoop Distributed File System
YARN = Yet Another Resource Negotiator
MapReduce = parallel processing model
```

Out[15]:
```
CompletedProcess(args='hdfs dfs -cat /user/lab/exercise1/file2.txt', ret
urncode=0, stdout='2026-09-13 23:47:40 WARN  NativeCodeLoader:60 - Unabl
e to load native-hadoop library for your platform... using builtin-java
classes where applicable\nFile 2: hadoop notes\nHDFS = Hadoop Distribute
d File System\nYARN = Yet Another Resource Negotiator\nMapReduce = paral
lel processing model\n', stderr='')
```

Exercise 5: Check the available space in HDFS

In [16]:
```
run("hdfs dfs -df -h /")
print()
run("hdfs dfsadmin -report")
```

```
$ hdfs dfs -df -h /
2026-09-13 23:47:41 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Filesystem                Size   Used  Available  Use%
hdfs://localhost:9000  460.4 G  4.7 K    113.3 G    0%

$ hdfs dfsadmin -report
2026-09-13 23:47:41 WARN  NativeCodeLoader:60 - Unable to load native-hado
op library for your platform... using builtin-java classes where applicabl
e
Configured Capacity: 494384795648 (460.43 GB)
Present Capacity: 121628251119 (113.28 GB)
DFS Remaining: 121628246016 (113.28 GB)
DFS Used: 5103 (4.98 KB)
DFS Used%: 0.00%
Replicated Blocks:
        Under replicated blocks: 0
        Blocks with corrupt replicas: 0
        Missing blocks: 0
        Missing blocks (with replication factor 1): 0
        Low redundancy blocks with highest priority to recover: 0
        Pending deletion blocks: 1
Erasure Coded Block Groups:
        Low redundancy block groups: 0
        Block groups with corrupt internal blocks: 0
        Missing block groups: 0
        Low redundancy blocks with highest priority to recover: 0
        Pending deletion blocks: 0

-------------------------------------------------
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: localhost
Decommission Status : Normal
Configured Capacity: 494384795648 (460.43 GB)
DFS Used: 5103 (4.98 KB)
Non DFS Used: 372756544529 (347.16 GB)
DFS Remaining: 121628246016 (113.28 GB)
DFS Used%: 0.00%
DFS Remaining%: 24.60%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sun Sep 13 23:47:41 IST 2026
Last Block Report: Sun Sep 13 23:47:23 IST 2026
Num of Blocks: 4
```

Out[16]:
```
CompletedProcess(args='hdfs dfsadmin -report', returncode=0, stdout='202
6-09-13 23:47:41 WARN  NativeCodeLoader:60 - Unable to load native-hadoo
p library for your platform... using builtin-java classes where applicab
le\nConfigured Capacity: 494384795648 (460.43 GB)\nPresent Capacity: 121
628251119 (113.28 GB)\nDFS Remaining: 121628246016 (113.28 GB)\nDFS Use
d: 5103 (4.98 KB)\nDFS Used%: 0.00%\nReplicated Blocks:\n\tUnder replica
ted blocks: 0\n\tBlocks with corrupt replicas: 0\n\tMissing blocks: 0\n
\tMissing blocks (with replication factor 1): 0\n\tLow redundancy blocks
with highest priority to recover: 0\n\tPending deletion blocks: 1\nErasu
re Coded Block Groups: \n\tLow redundancy block groups: 0\n\tBlock group
s with corrupt internal blocks: 0\n\tMissing block groups: 0\n\tLow redu
ndancy blocks with highest priority to recover: 0\n\tPending deletion bl
ocks: 0\n\n-------------------------------------------------\nLive datan
odes (1):\n\nName: 127.0.0.1:9866 (localhost)\nHostname: localhost\nDeco
mmission Status : Normal\nConfigured Capacity: 494384795648 (460.43 GB)
\nDFS Used: 5103 (4.98 KB)\nNon DFS Used: 372756544529 (347.16 GB)\nDFS
Remaining: 121628246016 (113.28 GB)\nDFS Used%: 0.00%\nDFS Remaining%: 2
4.60%\nConfigured Cache Capacity: 0 (0 B)\nCache Used: 0 (0 B)\nCache Re
maining: 0 (0 B)\nCache Used%: 100.00%\nCache Remaining%: 0.00%\nXceiver
s: 0\nLast contact: Sun Sep 13 23:47:41 IST 2026\nLast Block Report: Sun
Sep 13 23:47:23 IST 2026\nNum of Blocks: 4\n\n\n', stderr='')
```

Observation

- Java JDK and Hadoop were installed and configured for a single-node setup.
- Hadoop XML files (`core-site.xml` , `hdfs-site.xml`) were configured.
- The NameNode was formatted and HDFS daemons (NameNode, DataNode) started successfully.
- Directories were created in HDFS.
- Files were uploaded, listed, read, copied, downloaded, and deleted using HDFS shell commands.
- All exercise operations completed without errors.

Result

Hadoop was successfully configured, and all basic HDFS operations such as creating directories, uploading, listing, reading, downloading, and deleting files were performed successfully.

Conclusion

The experiment demonstrated the installation and configuration of Hadoop in a single-node environment. Basic HDFS operations were successfully performed using Hadoop shell commands, providing practical understanding of distributed storage and file management in Hadoop.